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OPTIMIZING HUMAN PERFORMANCE AT TSA AIRPORT SCREENING CHECKPOINTS: INTEGRATING PSYCHOLOGY IN SECURITY SYSTEMS

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Abstract – In today's persistent threat environment, strengthening the airport security screening checkpoint with its holistic human, social, and technological ecology in mind is an ongoing challenge. Research has focused on technology applications and the design of new or integrated systems, leaving out human-based solutions. A human factors psychology perspective can identify optimization potential without the need to commit to technology development. This paper discusses using human factors principles at airport security screening checkpoints to enhance security. The paper identifies three areas associated with human error at checkpoints. The focus is on how well system components (such as screening tasks) and organizational elements (such as scheduling/shiftwork, training, culture, communication, or procedures) are designed to interface with human physiological and cognitive abilities and limitations. This literature review focuses on the following critical elements in optimizing human performance at TSA airport security screening checkpoints: countering deception, preventing and mitigating fatigue, and fostering Critical Thinking and Effective Communication (CTEC).

Keywords: Human factors, security, system optimization, deception, fatigue

The terrorist attacks on the United States on September 11, 2001, occurred as terrorists overcame security screening procedures at Boston, Washington Dulles, and Newark airports (Kean & Hamilton, 2004). Subsequently, the Transportation Security Administration (TSA) was created as a component of the new U.S. Department of Homeland Security that was founded on November 25, 2002, and started operations on March 1, 2003. In today's environment, strengthening airport security screening checkpoints continues to be an important challenge (Janssen, van den Berg & Sharpanskykh, 2020; McFarlane, 2020; Walter, Schmutz & Grote, 2022). With increased demands for more agency funding and investment in technology for air transportation security (Berrick, 2008; Wang et al., 2014), it is also important to identify shortfalls and optimization potential related to human factor psychology. Such an additional focus promises security enhancement, with no need for large budget commitments and technology investment.

As the DHS Office of the Inspector General pointed out in a report titled "TSA needs to improve efforts to retain, hire, and train its Transportation Security Officers (TSO),"

> TSO responsibilities include operating various screening equipment and technology, performing pat-downs of passengers, conducting bag searches, controlling airport terminal entry and exit points, interacting with the public, and understanding and implementing TSA standard operating procedures [...]. In carrying out these screening responsibilities, TSOs must think critically about current threats and risks, and the responses necessary to mitigate

> those threats. (U.S. Department of Homeland Security 2019a, p.1)

However, the report found that training alone is not enough to develop the knowledge, skills, and abilities TSO's need to function optimally in their roles (U.S. Department of Homeland Security 2019a).

Against this challenge, this paper discusses the human factors principles that can be integrated into homeland security processes at airport security screening checkpoints. Human factors is an applied area of psychology that focuses on human-technology interactions (Stone et al., 2018). Human factors researchers use their knowledge of human capabilities to design technologies and systems that are best adapted to human capabilities. Discussing how human factors can be used to enhance the security screening process at airports is a relevant application of knowledge (Chartered Institute of Ergonomics and Human Factors, 2022). This work identified three main challenges at the TSA airport security screening checkpoint: countering deception, preventing and mitigating fatigue, and fostering critical thinking and effective communication (CTEC). The review identifies psychological principles relevant to security screening and TSA operations and suggests solutions to the issues identified.

Countering Deception at the TSA Screening Checkpoint

The first challenge in the TSA checkpoint system is detecting deception. Common sense would suggest that if detection of deception, and resultant threats related to deception, were countered early in the screening process, the air transportation system would be safer. Travelers carrying banned items such as weapons or dangerous substances would be prevented from entering the principal areas of airport terminals and from boarding flights. To make recommendations to enhance safety, understanding deception and detection is important.

Deception

Deception is more than just lying. The act of misleading or manipulating someone by false remarks, acts, or behavior with the goal of getting them to believe something untrue is referred to as deception (Galasiński, 2000; Granhag & Hartwig, 2012). It entails purposefully hiding or falsifying information to control people for one's own benefit, escape punishment, or accomplish a certain objective. Detection of deception is an important component of security screening, in order to ensure dangerous individuals or materials do not pass through checkpoints.

Galasiński's (2000) definition of deception is a good starting point for understanding deception. According to Galasiński, deception is defined as an act that is intended to foster in another person a belief or understanding that the deceiver considers false. Depending on the situation and the people involved, deception can take on different forms. However, there are several universal traits that are frequently linked to deceptive behavior. These consist of intentionality, false information, manipulation, motivation, and behavioral inconsistencies. Intent is a deliberate and intentional act of misleading others that constitutes deception. The deliberate effort to offer incorrect information is not accidental. Providing false information is a deceptive behavior that usually entails the dissemination of untrue information, either by blatant lying or by leaving out crucial details. Manipulation is when deceivers may employ a variety of strategies, such as delivering incomplete truths or causing distractions, to influence the perceptions or beliefs of their targets. There is typically a reason for motivation, which can range from self-interest or self-preservation to gaining a particular result or advantage. When it comes to inconsistent behavior, deceptive people may act inconsistently or give inconsistent verbal or body language cues while handling contradictory information.

Depending on the situation and the parties involved, deceit can serve a variety of purposes. Deception is typically used as a tactical strategy to accomplish specific objectives, acquire advantages, safeguard oneself, or control others. These could all be listed under the umbrella of personal gain. One can use deception to obtain personal advantages or benefits, such as financial gain, resource acquisition, or job advancement. In terms of self-preservation, people may use deception when they feel threatened or defenseless, to shield themselves from harm or punishment.

Indicators of Deception

By exploring empirical studies and theoretical frameworks, we can grasp the indicators that may manifest during deceptions. Two general types of deception indicators are behavioral and physiological (Vrij 2000; Vrij et al., 1996). The cues or symptoms used to determine if someone is being deceitful differ between behavioral and physiological markers of deception. Although both sorts of indicators are frequently investigated in the context of deception detection, they draw their data from various sources. Observable acts, verbal clues, and nonverbal behaviors that can indicate deceit are the focus of behavioral indicators (de Turck & Miller, 1985).

When someone is lying, their communication styles and body language frequently shift, which serves as one of these clues. Several typical behavioral signs of

deception include gaze aversion, speech errors, and nervousness (fidgeting). However, physiological indicators concentrate on alterations in the body's physiological reactions during deceitful behavior. These signs are frequently assessed using a range of physiological sensors and tools. Physiological signs of deception frequently include an increase in respiratory and heart rates, starting to sweat, one's mouth going dry, and a shaky voice.

Furthermore, indicators can be broken down into separate categories, which include those based in psychology and the study of human behavior (Cohn et al., 2007). Advocates of detection suggest that these psychological techniques can be used by many different professionals, such as security guards, counter-terrorism agents, and police interrogators. When psychological techniques are combined with vocal cues and speech patterns, it can lead to a 90% success rate when detecting deception (Adelson, 2004; Ekman & Friesen, 1978). Furthermore, the FACS, or Facial Action Coding System, is the coding system used to evaluate indicators (Cohn et al., 2007; Ekman & Rosenberg, 1997; Ekman & Friesen, 1978).

Psychological indicators refer to observable actions, facial expressions, and gestures displayed by individuals engaged in deception. There are three different theories and approaches regarding psychological indicators (Granhad & Hartwig, 2012). These include the Cognitive Complexity Theory (Spence, 2004; Spence et al., 2001), the Transactive Memory Approach (Driskell et al., 2012b) and the Emotional Deception Model (Gaspar & Schweitzer, 2013). All three of these have pros and cons in deception detection and should be used in certain situations.

The Cognitive Complexity Theory (Spence et al., 2000) is a cognition-based lie detection approach that seeks to understand and explain the cognitive processes underlying an individual's perceptions, problem-solving abilities, and decision-making skills. It argues that lying is a more cognitively demanding task than truth telling (Spence et al., 2001; Verschuere, Spruyt, Meijer, & Otgaar, 2011). It takes more mental power and thought to be able to suppress the pre-potent truth and appear innocent, especially during an in-person interview. People can be trained to use certain techniques that look for cognitive overload, such as asking people to tell their story backwards.

The Transactive Memory Approach requires two or more conspirators and depends on them being able to tell the correct story (Driskell et al., 2012a). Memories from one person may trigger memories of events in another person, leading them to fill in the story for each other. This means that people in continuing relationships develop a specialized division of labor with respect to encoding, storage, and retrieval of information. This approach is only valid when there are two or more suspects in question. Separating the suspects and asking unexpected questions to see if their story is lining up is a technique that can be easily trained for using this approach.

The most useful approach to deception detection, the Emotional Deception Model (EDM: Gaspar & Schweitzer, 2013), describes the three emotions associated with lying, and builds detection based on these emotions. This model is useful because it was developed and tested for use in applied settings, and it narrows the focus of the user to a small set of important emotions with a specific outcome. The three emphasized emotions are fear, guilt, and excitement. The focus of this approach is looking for leakage of these specific emotions. This can be shown through vocal and facial channels, such as elevated tone frequency and microexpressions. When looking for signs of heightened emotions, you can use techniques such as asking unexpected questions or asking individuals to participate in a secondary task during the interview. This could allow the brain to be focused on something else, with the resulting leakage causing individuals' true emotions to be visible through body movements, vocal cues, or microexpressions.

Elaborating on the EDM, the human voice is a potent instrument used for communication, but it may also provide important clues regarding sincerity (Anders Granhag & Vrij 2014; Newman, Pennebaker, Berry & Richards 2003). There are five main vocal indicators that can be used when trying to detect deception (Zuckerman et al., 1979). These include speech patterns, unnecessary pauses, restatement of the questions, avoiding "I" statements, and using the wrong words. During deceptive communication, individuals may exhibit changes in their speech patterns. These alterations can manifest as hesitations, inconsistencies, or shifts in their usual manner of speaking. For instance, a person may suddenly adopt a more formal or overly elaborate style to create a perceived distance from the lie. Deceptive individuals may also often use pauses or hesitation fillers (e.g., "um," "uh," or "you know") as they grapple with fabricating information on the spot. These pauses may be longer or more frequent than during truthful speech, providing potential clues to deception. Furthermore, when asked a direct question, deceptive individuals may rephrase or repeat the question before responding. This tactic serves as a stalling mechanism, allowing them time to construct a response that aligns with their deceptive narrative. People also tend to use first-person pronouns ("I," "me,"

or “my”) when recalling genuine experiences. However, when attempting to deceive, individuals might subconsciously avoid using "I" statements to create psychological distance from the lie, replacing them with third-person pronouns or passive constructions (e.g., "He said," "The document was found"). Finally, deceptive individuals may inadvertently use language that does not accurately convey the intended meaning or align with the context. These linguistic errors can include using incorrect verb tenses, mispronouncing words, or substituting one word for another that has a similar sound but a different meaning.

Despite technological advancements, human screeners remain essential in distinguishing between genuine anomalies in speech and true indications of deception, including in aviation security screening (Ormerod & Dando 2015; Driskell, Driskell, & Salas, E. 2012a). This is particularly important in moving from tactical to strategic detection of deception (Reid, Gozna & Boon 2016).

The EDM also affords the possibility of body/muscular indicators as a supplemental method to behavioral and verbal indicators in deception detection. This supplemental method is critically examined in this section (Ekman & Friesen, 1974). Attending to body/muscular indicators can lead to the successful detection of deception in many situations, increasing safety in situations (Gaspar & Schweitzer, 2013). During deceptive interactions, individuals may exhibit gaze aversion and other eye actions, indicative of deception (Gullipalli et al., 2021). These behaviors can stem from feelings of guilt, discomfort, or anxiety associated with the act of lying. Deceptive individuals may find it challenging to maintain eye contact, as they fear that their lies will be detected through their eyes.

An increase in blinking frequency may also show deception and is another body movement that has been associated with deception (Gullipalli et al., 2021). While it is natural for blinking rates to vary depending on environmental factors, increased blinking during a conversation or when responding to specific questions may indicate heightened stress or cognitive load associated with maintaining a deceptive narrative. Finally, foot and leg movements can reveal indicators of internal emotional states and deception (Vrij et al., 1996). For instance, fidgeting, tapping, or bouncing the legs may signify nervousness or discomfort during deceptive interactions. These subconscious movements can result from the stress of keeping up with a lie. It is important to note that body and muscular indicators should be interpreted in conjunction with other nonverbal cues, and the overall context of the situation. Some individuals may naturally exhibit certain body movements, and cultural norms can also influence nonverbal behavior.

Detection

Beyond understanding the elements of deception, it is important to understand if humans can detect the signs discussed above. Spain, Hedge & Blanchard (2017) examined human predictors of screening detection, such as personality traits, abilities, hobbies, and spatial abilities. The work showed that there was faster and more consistent detection with Transportation Security Officers versus Behavioral Detection Officers, but both were highly accurate. “Overall, the results of this analysis showed that visual search performance is strongly affected by behavioral factors, and that spatial ability played a lesser, but significant, role in search performance for TSOs. For BDOs [behavioral detection officers], video game play frequency also explained a significant amount of variance in visual search performance.” (Spain, Hedge & Blanchard, 2017 p.1311).

Non-professional individuals (those without TSA training) were faster at deception detection, however significantly less consistent with their results, than Transportation Security Officers (Biggs et al. 2013). The non-professional searchers completed a detection task with an overall lower accuracy. Furthermore, when comparing the experienced professionals in comparison to the early career professionals, there was a lower accuracy for the experienced professionals. Non-professionals also had a higher false alarm rate, and they took less time to locate the potential target in comparison to the professionals. The recommendation made by Biggs et al. (2013) is to emphasize consistency during professional visual search training and evaluation.

Deception Research Summary

It is critical to train screeners (security personnel, law enforcement, etc.) and frequent flyers (passengers who travel often and are familiar with the travel process) to spot deceptive behaviors. This is especially important in fields where deception detection is critical (e.g., the airport). Early threat detection can result in prompt intervention in potentially hazardous circumstances. This can stop crimes, disputes, or security breaches from getting worse. Regular travelers who have been taught to spot deceptive actions can be quite helpful in spotting suspicious people when they are traveling. If they observe someone acting in a suspicious manner, they can notify the appropriate authorities or airline personnel, potentially averting security breaches or incidents.

No single indicator of deception is foolproof, and reliance on a single cue may lead to inaccuracies and misinterpretations. Instead, a comprehensive approach,

combining various indicators and corroborating evidence, holds the key to enhancing the accuracy and reliability of deception detection processes. In conclusion, understanding the subtle intricacies of behavioral, vocal, and body/muscular indicators of deception can contribute to enhanced transportation safety. As humans strive to unravel the enigma of deception, this literature review serves as a foundation for future research endeavors that will enable people to navigate the delicate web of truth and falsehood with greater insight and accuracy.

Preventing and Mitigating Fatigue at the TSA Screening Checkpoint

The second area of focus for this review is fatigue. While there is no solid body of research specifically on fatigue as it affects Transportation Security Officers (TSOs) at TSA screening checkpoints at airports, the available research on air traffic controllers and fatigue provides transferable insight.

Fatigue significantly impacts air traffic controllers, affecting their physical, mental, and emotional well-being, which in turn compromises their work performance. Cognitive functions such as concentration, dexterity, and intellectual processing are impaired, leading to an increased potential for workplace errors and a compromise in safety (Ferreira et al., 2020). In the high-stakes environment of air traffic control, precision and quick decision-making are imperative, and the balancing act between speed and accuracy becomes crucial. Similarly, the relevance of the speed-accuracy tradeoff at screening checkpoints is important. Accumulating sufficient evidence for decisions enhances accuracy but may slow response times, while faster responses can result in inaccuracies (Donkin et al., 2014). This delicate balance between speed and accuracy is further complicated by the nature of shift work common in these fields, which can disrupt circadian rhythms and sleep quality, exacerbate fatigue, and increase the cognitive workload (Takeyama et al., 2005; Zhang et al., 2021).

Fatigue is typically studied in three components: physical, mental, and emotional (Kop & Kupper, 2016). Physical fatigue, which may stem from sleep deprivation, over-exertion, illness, or inadequate nutrition, can significantly impact job satisfaction. For TSA workers, diminished job satisfaction can hamper performance, which can lead to decreased personal achievements and adverse organizational outcomes. These outcomes can manifest as counterproductive work behavior, increased turnover intentions, and higher incidences of illness (Baeriswyl et al., 2016). These behaviors could result in decreased workplace safety as well as air transport security gaps.

Mental fatigue promotes a decrease in motivation, feelings of exhaustion, lack of energy, and impairs cognitive function and plays a key role in increasing error risk in the workplace. It alters the amount of physical effort people want to invest and is associated with a reduction in the involvement of the prefrontal cognitive control processes (Ferreira et al. 2020).

Working under challenging conditions may precipitate emotional fatigue, which Baeriswyl et al. (2016) described as exhaustion due to the intense physical and mental exertions demanded by high-stress jobs. Such a state typically presents diminished work performance and can potentially lead to burnout, characterized by a diminished sense of personal identity and a reduced capacity to fulfill work responsibilities (Barnes & Van Dyne, 2009).

Fatigue increases tradeoffs, particularly related to speed vs. accuracy. Collecting more evidence before deciding allows one to be more accurate but slower (Donkin et al., 2014). The primary influencer of caution is the threshold amount of evidence required to trigger a response. People will ignore redundant information when they are forced to respond quickly even when information is consistently repeated or reinforced. The primary influencer of the speed-accuracy tradeoff is not the capacity of a processing system, but the amount of evidence required to make a choice. Donkin et al. (2014) reported that participants were capable of strategically ignoring redundant information when being forced to process information quickly: participants did not collect less evidence before making a quick response, rather they found ways to reduce the load of their processing system.

Preventing and Mitigating Fatigue

There are several ways in which fatigue can be mitigated that are applicable to the security screening environment. Aligning work schedules with an individual's chronotype, which is an individual's preferred times across the 24-hour cycle of activity and rest, has been shown to mitigate the effects of fatigue, reducing mental workload, and enhancing performance (Venesia et al., 2019). Additionally, collaborative efforts among TSOs to align schedules based on chronotype have proven more effective than reliance on technology alone, emphasizing the vital role of human factors in maintaining safety and efficiency (Yamani et al., 2018).

Addressing psychological stress is also crucial, as it influences fatigue, resilient behavior, and workplace safety. Implementing recovery processes and managing stressors can lead to a more positive and resilient safety

culture, ensuring well-being and optimal performance. This has been shown for air traffic controllers (Langan-Fox et al., 2009) and is a finding also relevant for human factors-based TSA airport screening checkpoint optimization efforts.

Ultimately, tackling fatigue in air transportation security demands a comprehensive approach, incorporating schedule adjustments, an understanding of the speed-accuracy tradeoff, collaborative work strategies, and proactive stress management.

Fostering Critical Thinking and Effective Communication (CTEC) at the TSA Screening Checkpoint

Critical thinking is the cornerstone of a TSO's decision-making process, enabling them to analyze complex situations, anticipate potential threats, and make precise decisions. Effective communication ensures that critical information is accurately conveyed among team members, stakeholders, and passengers, mitigating the risk of human error (Aydiner, 2022).

The integration of critical thinking and effective communication (CTEC) within TSA training programs faces obstacles, as highlighted by a Department of Homeland Security report that underscored the challenges in retention, training, and recruitment, attributing them partly to deficiencies in CTEC competencies among TSA leaders (U.S. Department of Homeland Security, 2019a). Such internal findings indicate a significant gap in current training frameworks and emphasize the need for strategic enhancements. There are several strategies for CTEC enhancement, as summarized in the following.

- Red Teaming Techniques: Red teaming involves creating an internal or external team to simulate the thoughts or behaviors of your adversaries. This practice finds weaknesses in defenses, and can challenge flawed responses (TechTarget, 2024). Adopting red teaming offers a structured approach to challenging assumptions and enhancing decision-making. This technique allows TSA to simulate adversarial scenarios, thereby improving preparedness for real-world threats (United Kingdom Ministry of Defense, 2021; U.S. Army Combined Arms Center, 2021).
- Inquiry-Based Learning: Encouraging TSA personnel to engage in critical questioning fosters a deeper understanding of security scenarios. This approach cultivates adaptive problem-solving, and a mindset geared towards continuous improvement (Paul & Elder, 2000).
- Leveraging FEMA's Communication Training: FEMA's communication training materials provide valuable resources for developing effective communication skills within disaster and emergency contexts, crucial for maintaining clear communication in critical situations (Kirschenbaum & Rapaport, 2017; United States Federal Emergency Management Agency, 2021).

In conclusion, integrating CTEC into TSA's training and operational paradigms is essential for advancing aviation security (Kiltz, 2009). Embedding critical thinking and effective communication at the core of TSA's culture enhances the agency's ability to respond to threats with agility and precision. This endeavor requires concerted effort across training, policy, and operational domains (Aydiner, 2022; Department of Homeland Security, 2019a; United Kingdom Ministry of Defense, 2021; Paul & Elder, 2005; 2000; United States Federal Emergency Management Agency, 2021).

Conclusion and Summary of Recommendations

Research in the homeland security discipline has focused primarily on technology applications and better design of new or integrated systems to enhance safety, rather than behavioral optimization (Baggett et al., 2017). This paper proposed three areas in which psychology could provide an integrated approach to transportation security (Neffenger & Ades, 2020). Studying the optimization of human performance in the air transportation sector of the homeland security enterprise relates to several core DHS missions including: preventing terrorism, securing critical infrastructure, ensuring resilience to disasters, and growing the homeland security workforce (Siedschlag & Jerković, 2022).

Previous studies focused on airport security screeners' knowledge related to problem-solving during screening tasks (e.g., Swann et al., 2020) but did not address multi-modal screening (such as x-raying combined with behavioral observation in the social setting of the checkpoint and its surroundings). One area where more work can be done is integrating deception detection training into the overall screener training paradigm.

There has been a plethora of research into both visual search tasks and object monitoring tasks as they relate to TSA officers and air traffic controllers, but a literature search showed much less research into the effects of sleep deprivation from shift work on performance. For example, there are differences between TSA officers and nonprofessional searchers. TSA professionals are more accurate than nonprofessionals, but also slower (Biggs et al., 2013; Biggs et al., 2018). Could a better understanding of how fatigue contributes to screener performance be used to modify schedules of

workers, so that screeners can enhance speed of scanning without sacrificing accuracy?

The present work also found that transportation security may be hampered by communication difficulties across the DHS organization, as well as lack of training, especially related to critical thinking and decision-making. A report by the Office of Inspector General of the U.S. Department of Homeland Security, titled "Major Management and Performance Challenges Facing the Department of Homeland Security" (Office of Inspector General, 2022), suggested mismatches between technology and users, and communication/collaboration lapses between multiple agencies thus providing an interesting relevant starting point for integrating human factors psychology knowledge into the air transportation system. How can psychological principles be used by DHS to enhance understanding of decision-making and when/how information is best shared across relevant parties. TSA supports the need for training by calling for research on how to improve operational performance at the screening checkpoint with the human user in mind, thus improving efficiency, effectiveness, and public safety (Kudrick & Greene, 2021).

This review argued for an approach that addresses human performance optimization in the air transportation system by integrating human factors psychology principles into homeland security deterrence and detection tasks, procedures, and training (Biggs et al., 2018). The focus was on how well system components (such as screening tasks) and organizational elements (such as scheduling/shiftwork, training, culture, communication, or procedures) are designed to interface with human physiological and cognitive abilities and limitations.

References

Adelson, R. (2004, July 1). Psychological sleuths--Detecting deception. *Monitor on Psychology*, *35*(7). https://www.apa.org/monitor/julaug04/detecting

Anders Granhag, P., & Vrij, A. (2014). Interviewing to detect deception. *Interviewing to Detect Deception, 19(3)*, 75–93. https://doi.org/10.4324/9781315094038-5

Aydiner C. (2022). Critical thinking and effective communication in security domains. In: Masys, A.J. (eds) *Handbook of Security Science*. Springer, Cham. https://doi.org/10.1007/978-3-319-91875-4_2

Baeriswyl, S., Krause, A., & Schwaninger, A. (2016). Emotional exhaustion and job satisfaction in airport security officers - work-family conflict as mediator in the job demands-resources model. *Frontiers in Psychology*, *7*, 663. https://doi.org/10.3389/fpsyg.2016.00663

Baggett, R. K., Foster, C. S., & Simpkins, B. K. (Eds.) (2017). *Homeland security technologies for the 21st century*. Praeger. http://publisher.abc-clio.com/9781440831430

Barnes, C. M., & Van Dyne, L. (2009). `I'm tired': Differential effects of physical and emotional fatigue on workload management strategies. *Human Relations, 62*(1), 59-92. https://doi.org/10.1177/0018726708099518

Berrick, C. A. (2008). *Transportation security: Transportation security administration has strengthened planning to guide investments in key aviation and surface transportation security programs, but more work remains: Congressional testimony*. DIANE Publishing. https://www.govinfo.gov/content/pkg/GAOREPORTS-GAO-08-456T/html/GAOREPORTS-GAO-08-456T.htm

Biggs, A. T., Cain, M. S., Clark K., Darling, E. F., & Mitroff, S. R. (2013). Assessing visual search performance differences between Transportation Security Administration Officers and nonprofessional visual searchers. *Visual Cognition, 21(*3), 330-352. https://doi.org/10.1080/13506285.2013.790329

Biggs, A. T., Kramer, M. R., & Mitroff, S. R. (2018). Using cognitive psychology research to inform professional visual search operations. *Journal of Applied Research in Memory and Cognition*, *7*(2), 189-198. https://doi.org/10.1016/j.jarmac.2018.04.001

Chartered Institute of Ergonomics and Human Factors (2022). Human factors in highly automated systems. https://ergonomics.org.uk/asset/B1F30FC8-5E44-4610-B19122A70B1735F7

Cohn, J. F., Ambadar, Z., & Ekman, P. (2007). Observer-based measurement of facial expression with the Facial Action Coding System. *The Handbook of Emotion Elicitation and Assessment*, *1*(3), 203-221. https://www.proquest.com/psycinfo/books/observer-based-measurement-facial-expression-with/docview/621779879/sem-2?accountid=27203

de Turck, M. A., & Miller, G. R. (1985). Deception and arousal: Isolating the behavioral correlates of deception. *Human Communication Research*, *12*(2), 181-201. https://doi.org/10.1111/j.1468-2958.1985.tb00072.x

Donkin, C., Little, D. R., & Houpt, J. W. (2014). Assessing the speed-accuracy trade-off effect on the capacity of information processing. *Journal of Experimental Psychology: Human Perception*

and Performance, 40(3), 1183-1202. https://doi.org/10.1037/a0035947

Driskell, T. E., Driskell, J. E., & Salas, E. (2012). Social indicators of deception: Training to detect deception in conspirators. *Social Indicators of Deception*. https://doi.org/10.1037/e642172012-001

Driskell, J. E., Salas, E., & Driskell, T. (2012b). Social indicators of deception. *Human Factors*, *54*(4), 577-588. https://doi.org/10.1177/0018720812446338

Ekman, P., & Rosenberg, E. L. (Eds.). (1997). *What the face reveals: Basic and applied studies of spontaneous expression using the Facial Action Coding System (FACS)*. Oxford University Press, USA.https://psycnet.apa.org/doi/10.1093/acprof:oso/9780195179644.001.0001

Ekman, P., & Friesen, W. V. (1978). Facial action coding system. *Environmental Psychology & Nonverbal Behavior*. https://psycnet.apa.org/doi/10.1037/t27734-000

Ekman, P., & Friesen, W. V. (1974). Detecting deception from the body or face. *Journal of personality and Social Psychology*, *29*(3), 288. https://psycnet.apa.org/doi/10.1037/h0036006

Ferreira, A., Jr., Chierotti, P., Gabardo, J. M., Giovanini, B., Okano, A. H., Buzzachera, C. F., Okazaki, V. H. A., Okuno, N. M., & Altimari, L. R. (2020). Residual effects of mental fatigue on subjective fatigue, reaction time and cardiac responses. *Revista De Psicología Del Deporte, 29*(2), 27-34. http://ezproxy.libproxy.db.erau.edu/login?url=https://www.proquest.com/scholarly-journals/residual-effects-mental-fatigue-on-subjective/docview/2594716411/se-2

Galasiński, D. (2000). The language of deception: A discourse analytical study. Sage: Thousand Oaks, CA. https://onlinelibrary.wiley.com/doi/abs/10.1002/acp.765

Gaspar, J. P., & Schweitzer, M. E. (2013). The emotion deception model: A review of deception in negotiation and the role of emotion in deception. *Negotiation and Conflict Management Research*, *6*(3), 160-179. https://doi.org/10.1111/ncmr.12010

Granhag, P. A., & Hartwig, M. (2012). Detecting deception. In Davies & Beech (Eds.) *Forensic Psychology: Crime, Justice, Law, Interventions* (pp. 151-169). John Wiley & Sons. https://awspntest.apa.org/doi/10.1037/14462-008

Gullapalli, A. R., Anderson, N. E., Yerramsetty, R., Harenski, C. L., & Kiehl, K. A. (2021). In the blink of an eye: Quantitative blink dynamics predict deceptive personality traits in forensic interviews. *Personality and Individual Differences*, *176*, 110764. https://doi.org/10.1016/j.paid.2021.110764

Janssen, S., van den Berg, A., & Sharpanskykh, A. (2020). Agent-based vulnerability assessment at airport security checkpoints: A case study on security operator behavior. *Transportation Research Interdisciplinary Perspectives, 5,* 100139. https://doi.org/10.1016/j.trip.2020.100139.

Kean, T. H., & Hamilton, L. (2004). *The 9/11 commission report: Final report of the national commission on terrorist attacks upon the United States* (Vol. 1). Government Printing Office. Retrieved from https://9-11commission.gov/report/911Report.pdf

Kiltz L (2009) Developing critical thinking skills in homeland security and emergency management courses. *Journal of Homeland Security Emergency Management, 6*(1):1–25. https://doi.org/10.2202/1547-7355.1558

Kirschenbaum, A., & Rapaport, C. (2017). Does training improve security decisions? A case study of airports. *Security Journal*, *30*, 184-198. https://doi.org/10.1057/sj.2014.39

Kop, W. J., & Kupper, H. M. (2016). Fatigue and stress. In *Stress: Concepts, Cognition, Emotion, and Behavior* (pp. 345-350). Academic Press. https://www.proquest.com/psycinfo/books/fatigue-stress/docview/1817572117/sem-2?accountid=27203

Kudrick, B. & Greene, F. (2021). Human factors engineering at the transportation security administration [Paper presentation]. Applied Human Factors and Ergonomics International. https://openaccess.cms-conferences.org/#/publications/book/978-1-4951-2097-8

Langan-Fox, J., Sankey, M.J. & Canty, J.M. (2009). Human factors measurement for future air traffic control systems. *Human Factors, 51*(5), 595-637. https://doi.org/10.1177/0018720809355278

McFarlane, P. (2020). Linking aviation security failures to human-mediated error. A review of the related literatures with directions for policy and research. *Journal of Transportation Security 13,* 33-51. https://doi.org/10.1007/s12198-020-00209-z

Neffenger, P. & Ades, R. (2020). Rethinking transportation security. In Bersin, A., Lawson, C., & Kayyem, J. N. (Eds.): *Beyond 9/11: Homeland Security for the Twenty-first Century*. Belfer Center for Science and International Affairs. https://doi.org/10.7551/mitpress/13831.003.0009

Newman, M. L., Pennebaker, J. W., Berry, D. S., & Richards, J. M. (2003a). Lying words: Predicting deception from linguistic styles. *Personality and Social Psychology Bulletin*, *29*(5), 665–675. https://doi.org/10.1177/0146167203029005010

Office of Inspector General (2022). Major management and performance challenges facing the Department of Homeland Security. Washington, D.C.: U.S. Department of Homeland Security. https://www.oig.dhs.gov/sites/default/files/assets/2022-11/OIG-23-01-Oct22.pdf

Ormerod, T. C., & Dando, C. J. (2015). Finding a needle in a haystack: Toward a psychologically informed method for aviation security screening. *Journal of Experimental Psychology: General*, *144*(1), 76–84. https://doi.org/10.1037/xge0000030

Paul, R. & Elder, L. (2000) *Critical thinking handbook: Basic theory and instructional structures*. http://www.criticalthinking.org.resources/articles/therole-of-questions.html.

Paul, R. & Elder, L. (2005) *The miniature guide to critical thinking: concepts and tools*. The Foundation for Critical Thinking. https://www.criticalthinking.org/files/Concepts_Tools.pdf.

Reid, I. D., Gozna, L. F., & Boon, J.C.W. (2016). From tactical to strategic deception detection: Application of psychological synthesis. *Journal of Strategic Security 10*(1) 81-101. http://doi.org/10.5038/1944-0472.10.1.1528

Siedschlag, A. & Jerković, A. (2022). Homeland and civil security research studies for an evolving mission space. In A. Siedschlag and A. Jerković (Eds.), *Cross-disciplinary Perspectives on Homeland and Civil Security: A Research-based Introduction* (pp. 1-35). Peter Lang. https://doi.org/10.3726/b19109

Spain, R. D., Hedge, J. W., & Blanchard, J. K. (2017). Examining predictors of visual search success in transportation security officers and behavior detection officers. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *61*(1), 1308–1312. https://doi.org/10.1177/1541931213601809

Spence, S. A. (2004). The deceptive brain. *Journal of the Royal Society of Medicine*, *97*(1), 6-9. https://doi.org/10.1177/014107680409700133

Spence, S. A., Farrow, T. F., Herford, A. E., Wilkinson, I. D., Zheng, Y., & Woodruff, P. W. (2001). Behavioural and functional anatomical correlates of deception in humans. *Neuroreport*, *12*(13), 2849-2853. https://doi.org/10.1097/00001756-200109170-00019

Stone, N., Chaparro, A., Keebler, J., Chaparro, B., & McConnell, D. (2018). *Introduction to human factors: Applying psychology to design*. CRC Press: Boca Raton, FL. https://doi.org/10.1201/9781315153704

Swann, L., Popovic, V., Blackler, A., & Thompson, H. (2020). Airport security screener problem-solving knowledge and implications. *Human Factors, 62*(8), 1265-1285. https://doi.org/10.1177/0018720819874169

Takeyama, H., Itani, T., Tachi, N., Sakamura, O., Murata K., Inoue T., Takanishi T., Suzumura H., & Niwa S., (2005) Effects of shift schedules on fatigue and physiological functions among firefighters during night duty, *Ergonomics, 48*:1, 1-11. https://doi.org/10.1080/00140130412331303920

TechTarget (2024). What is red teaming? https://www.techtarget.com/whatis/definition/red-teaming

United Kingdom Ministry of Defense (MOD) (2021) Red teaming guide (3rd ed.)https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1027158/20210625-Red_Teaming_Handbook.pdf

U.S. Federal Emergency Management Agency (FEMA) (2021) IS-242.C: Effective communication instructor guide. https://training.fema.gov/is/courseoverview.aspx?code=IS-242.c&lang=en

U.S. Department of Homeland Security, Office of the Inspector General. (2019a, March 28). TSA needs to improve efforts to retain, hire, and train its Transportation Security Officers. https://www.oig.dhs.gov/sites/default/files/assets/2019-04/OIG-19-35-Mar19.pdf

U.S. Army Combined Arms Center (2021). Thinking handbook ver. 9.0. UFMCS, Fort Leavenworth. https://usacac.army.mil/sites/default/files/documents/ufmcs/The_Red_Team_Handbook.pdf

Venesia, Z., & Widyanti, A. (2019, March). Compatibility between shift work and chronotype in Indonesian air traffic control workers: View of performance and mental workload. In *Journal of Physics: Conference Series, 1175*(1), 012191. Retrieved from https://iopscience.iop.org/article/10.1088/1742-6596/1175/1/012191/pdf

Verschuere, B., Spruyt, A., Meijer, E. H., & Otgaar, H. (2011). The ease of lying. *Consciousness and Cognition*, *20*(3), 908-911. https://doi.org/10.1016/j.concog.2010.10.023

Vrij, A., Semin, G. R., & Bull, R. (1996). Insight into behavior displayed during deception. *Human Communication Research*, *22*(4), 544-562. https://doi.org/10.1111/j.1468-2958.1996.tb00378.x

Vrij, A. (2000). *Detecting lies and deceit: The psychology of lying and the implications for professional practice*. Wiley: Chichester. https://onlinelibrary.wiley.com/doi/abs/10.1002/casp.828

Walter, S., Schmutz, J. B., & Grote, G. (2022). A safety approach for improving security: effective coordination strategies at the airport security screening checkpoint. *Human Factors, 66*(4),

1302-1314. https://doi.org/10.1177/0018720822112141 1

Wang, Y., Kofi Andoh-Baidoo, F. & Sun, J. (2014). Security investment in the aviation industry: A longitudinal analysis. *Industrial Management & Data Systems*, *114*(2), 276-291. https://doi.org/10.1108/IMDS-04-2013-0176

Yamani, Y., & Horrey, W. J. (2018). A theoretical model of human-automation interaction grounded in resource allocation policy during automated driving. *International Journal of Human Factors and Ergonomics*, *5*(3), 225-239. https://doi.org/10.1504/IJHFE.2018.095912

Zhang, J., Chen, Z., Liu, W., Ding, P., & Wu, Q. (2021). A field study of work type influence on air traffic controllers' fatigue based on data-driven PERCLOS detection. *International Journal of Environmental Research and Public Health*, *18*(22), 11937. http://dx.doi.org/10.3390/ijerph182211937

Zuckerman, M., DeFrank, R. S., Hall, J. A., Larrance, D. T., & Rosenthal, R. (1979). Facial and vocal cues of deception and honesty. *Journal of Experimental Social Psychology*, *15*(4), 378-396. https://doi.org/10.1016/0022-1031(79)90045-3

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